

MAT21-4

# Discrete Stochastic Models & Concentration

Conditional expectation, random walks, Markov chains, concentration inequalities

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Structured stochastic reasoning and quantitative uncertainty bounds on finite and countable spaces.

Albert School — 22 hours

# Preface

You arrive at this course already fluent in three languages. From *Discrete Mathematics* (MAT11-4) you know how to slice a set into an **equivalence partition** and reason case by case — “given that we are in block  $A$ , what fraction of  $A$  has property  $B$ ?” From *Probability & Statistics I* (MAT21-2) you know random variables, expectation, variance, the classic discrete laws, and the total-probability formula. From *Linear Algebra I* (MAT11-3) you know how to multiply matrices and raise them to powers, and you have already met finite state transitions as matrices.

This course fuses the three. **Conditional expectation** is nothing but the weighted-average-over-a-partition idea of MAT11-4, now applied to a random variable instead of a count. **Markov chains** are the matrix powers of MAT11-3, now read as probabilities. And the new ingredient — **concentration** — answers the question MAT21-2 raised but never closed: knowing the mean and variance of an estimator, *how many observations* do we actually need before we may trust it? We will keep the apparatus light: no sigma-algebras, no formal filtrations. Those arrive, built from scratch, in S3 and S6. Here every idea is introduced on a finite or countable space where you can, in principle, list every outcome.

Each chapter opens with a concrete example you can compute by hand, then states the general principle, then flags the mistakes students reliably make. Read with a pen.

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# 1 Conditional expectation on finite partitions

## 1.1 From “average over a block” to a random variable

**Worked example — A warm-up you can already do.** Roll a fair die,  $\Omega = \{1, \dots, 6\}$ , and let  $X(\omega) = \omega$  be the face. A friend, without showing you the die, tells you only whether the result is *low* ( $\in \{1, 2, 3\}$ ) or *high* ( $\in \{4, 5, 6\}$ ). What is your best guess for  $X$ ?

If told “low”, you average 1, 2, 3 to get 2. If told “high”, you average 4, 5, 6 to get 5. Your guess is therefore a *new random variable*: it equals 2 on the block  $\{1, 2, 3\}$  and 5 on the block  $\{4, 5, 6\}$ . That variable is the conditional expectation  $E[X \mid \mathcal{P}]$ . It is the heart of this chapter.

We work on a finite probability space  $(\Omega, P)$  and let  $X : \Omega \rightarrow \mathbb{R}$  be a random variable. A **partition** of  $\Omega$  is a family  $\mathcal{P} = \{B_1, \dots, B_k\}$  of non-empty disjoint events with  $\bigcup_i B_i = \Omega$  and  $P(B_i) > 0$ . Exactly as in MAT11-4, a partition encodes a piece of information: **which block occurred**.

**Definition 1 (Conditional expectation given a block)** For an event  $B$  with  $P(B) > 0$ ,

$$E[X \mid B] = \sum_x x P(X = x \mid B) = \frac{1}{P(B)} \sum_{\omega \in B} X(\omega) P(\omega).$$

It is the ordinary expectation of  $X$ , but re-weighted so that the block  $B$  now carries all the probability.

**Definition 2 (Conditional expectation given a partition)** The conditional expectation of  $X$  given  $\mathcal{P} = \{B_1, \dots, B_k\}$  is the random variable  $E[X \mid \mathcal{P}]$  that is **constant on each block**, taking the value  $E[X \mid B_i]$  throughout  $B_i$ :

$$E[X \mid \mathcal{P}](\omega) = E[X \mid B_i] \quad \text{if } \omega \in B_i.$$

The key mental shift — and the one students find hardest — is that  $E[X \mid \mathcal{P}]$  is *not a number* but a *function of the outcome*. It is the coarsened, “blurred” version of  $X$  you would report if you could see only which block occurred.

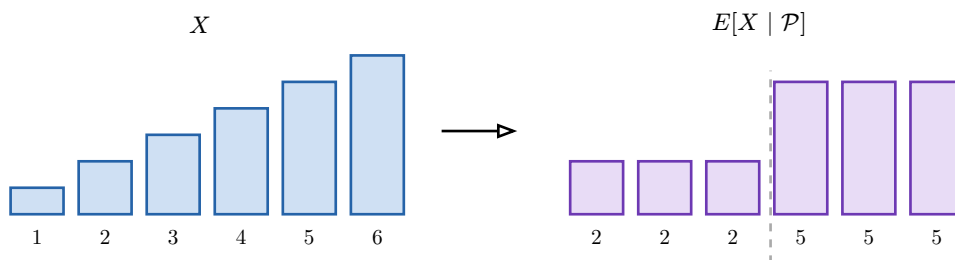


Figure 1: Conditioning on the low/high partition replaces the six distinct values of  $X$  (left) by the block averages 2 and 5 (right): a coarser random variable, constant on each block.

**Remark** The block-averaging always preserves the overall mean:  $E[E[X \mid \mathcal{P}]] = E[X]$ . In the die example,  $\frac{1}{2} \cdot 2 + \frac{1}{2} \cdot 5 = 3.5 = E[X]$ . We promote this observation to a theorem in the next chapter.

## 1.2 Partition refinement and the tower property

Computation proceeds by **partition refinement**: to evaluate  $E[X \mid \mathcal{P}]$  you restrict to one block at a time and average  $X$  over it. Refinement is also the right language for *gaining information in stages*.

**Definition 3 (Refinement)** A partition  $\mathcal{Q}$  **refines**  $\mathcal{P}$  (written  $\mathcal{Q} \preceq \mathcal{P}$ ) if every block of  $\mathcal{Q}$  is contained in some block of  $\mathcal{P}$ . Then  $\mathcal{Q}$  is *finer* (more informative) and  $\mathcal{P}$  is *coarser*.

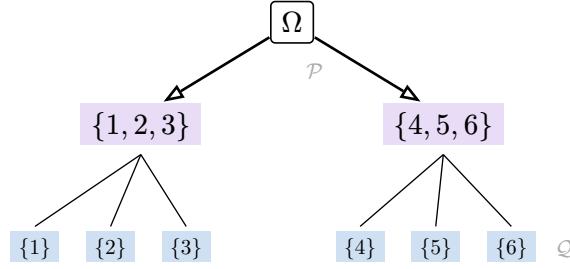


Figure 2: The singleton partition  $\mathcal{Q}$  (bottom) refines the low/high partition  $\mathcal{P}$  (middle), which in turn refines the trivial partition  $\{\Omega\}$  (top). Information increases downward.

**Theorem 4 (Tower property)** If  $\mathcal{Q}$  refines  $\mathcal{P}$ , then

$$E[E[X \mid \mathcal{Q}] \mid \mathcal{P}] = E[X \mid \mathcal{P}].$$

In particular, taking  $\mathcal{P} = \{\Omega\}$ ,

$$E[E[X \mid \mathcal{Q}]] = E[X].$$

Read it as a *consistency of averaging*: averaging a fine-grained guess back up to a coarse partition returns exactly the coarse guess. You cannot create information by first refining and then re-coarsening.

**Worked example — Verifying the tower property on the die.** Take  $\mathcal{Q} =$  singletons (so  $E[X \mid \mathcal{Q}] = X$ , since each block pins down  $\omega$ ) and  $\mathcal{P} = \{\{1, 2, 3\}, \{4, 5, 6\}\}$ . Then  $E[E[X \mid \mathcal{Q}] \mid \mathcal{P}] = E[X \mid \mathcal{P}]$ , which is 2 on the low block and 5 on the high block — the same variable we built by hand in the warm-up.

**Common pitfall** Three recurring mistakes.

- **Treating  $E[X \mid \mathcal{P}]$  as a number.** It is a random variable; only  $E[X \mid B_i]$  for a single block is a number.
- **Inverting the tower.** The inner conditioning must be on the *finer* partition:  $E[E[X \mid \mathcal{Q}] \mid \mathcal{P}]$  collapses to  $E[X \mid \mathcal{P}]$ , but  $E[E[X \mid \mathcal{P}] \mid \mathcal{Q}]$  does *not* simplify the same way.
- **Forgetting the weights.**  $E[X] = \sum_i E[X \mid B_i] P(B_i)$ , not the plain average of the block means — unless the blocks are equiprobable.

## 2 Variance decomposition: explained vs. residual

### 2.1 The two questions a partition answers

Conditioning on a partition asks two questions at once. The law of total *expectation* asks: **on average, what is  $X$ ?** The law of total *variance* asks: **how much of  $X$ 's variability does the partition explain, and how much remains inside the blocks?** This “explained vs. residual” split is the same idea you will later meet as  $R^2$  in regression.

**Theorem 5 (Law of total expectation)** For a partition  $\mathcal{P} = \{B_1, \dots, B_k\}$ ,

$$E[X] = \sum_{i=1}^k E[X \mid B_i] P(B_i) = E[E[X \mid \mathcal{P}]].$$

**Theorem 6 (Law of total variance)**

$$\text{Var}(X) = \underbrace{E[\text{Var}(X \mid \mathcal{P})]}_{\text{residual: within-block}} + \underbrace{\text{Var}(E[X \mid \mathcal{P}])}_{\text{explained: between-block}}.$$

Here  $\text{Var}(X \mid \mathcal{P})$  is the random variable equal to the within-block variance  $\text{Var}(X \mid B_i)$  on  $B_i$ ; its expectation is the average spread that survives *after* you learn the block. The second term measures how far the block means scatter around the grand mean — the variability the partition *accounts for*.

**Worked example — Two coins — splitting the variance.** Pick one of two coins uniformly: coin  $L$  lands heads with probability 0.2, coin  $H$  with probability 0.8. Flip the chosen coin once and let  $X \in \{0, 1\}$  be the result; let  $\mathcal{P}$  record which coin was picked.

- **Residual.** Each coin alone has variance  $p(1-p) = 0.2 \cdot 0.8 = 0.16$ , so  $E[\text{Var}(X \mid \mathcal{P})] = 0.16$ .
- **Explained.** The block means are 0.2 and 0.8 around the grand mean 0.5, giving  $\text{Var}(E[X \mid \mathcal{P}]) = \frac{1}{2}(0.2 - 0.5)^2 + \frac{1}{2}(0.8 - 0.5)^2 = 0.09$ .
- **Total.**  $\text{Var}(X) = 0.16 + 0.09 = 0.25$ , matching  $0.5 \cdot 0.5$  directly.

Knowing *which coin* explains  $\frac{0.09}{0.25} = 36\%$  of the variability; the rest is the irreducible randomness of a single flip.

**Common pitfall** The residual term is  $E[\text{Var}(X \mid \mathcal{P})]$  — the *expected* within-block variance, averaged over blocks with their probabilities — not the variance of any one block, and not  $\text{Var}(E[\text{Var}(X \mid \mathcal{P})])$ . Mixing up which term gets the outer expectation and which gets the outer variance is the single most common slip here.

## 3 Martingale intuition: the fair game

### 3.1 Betting without an edge

**Worked example — A fair coin game.** You bet on fair coin flips, winning or losing one euro each round. Let  $X_n$  be your fortune after  $n$  rounds. Whatever has happened so far, your expected fortune after the next flip is exactly what you hold now: the game has no drift. A

sequence with this “no expected change” property is a **martingale** — the mathematical model of a *fair game*.

**Definition 7 (Martingale (fair game))** A sequence  $X_0, X_1, X_2, \dots$  is a martingale if for every  $n$ ,

$$E[X_{n+1} \mid X_0, \dots, X_n] = X_n.$$

The conditional expected next value equals the current value: on average the game neither gains nor loses. (We condition on the observed history  $X_0, \dots, X_n$  informally, exactly as the partition “which path so far” — no filtration formalism is needed at this level.)

Two canonical examples carry us through the rest of the course.

- **Symmetric random walk:**  $S_n = \sum_{i=1}^n \xi_i$  with  $\xi_i = \pm 1$  equally likely and independent. Since  $E[\xi_{n+1}] = 0$  independently of the past,  $E[S_{n+1} \mid S_0, \dots, S_n] = S_n + E[\xi_{n+1}] = S_n$ .
- **Doob martingale:** fix a target random variable  $Y$  and a refining sequence of partitions  $\mathcal{P}_0 \not\preceq \dots$ ; set  $X_n = E[Y \mid \mathcal{P}_n]$ . The tower property makes  $(X_n)$  a martingale: your best estimate of  $Y$  updates, on average, to itself as information accrues.

### 3.2 Optional stopping and where it breaks

A **stopping time**  $T$  is a random time you can recognise as it happens: whether “ $T \leq n$ ” holds depends only on  $X_0, \dots, X_n$ , not on the future. “Stop at the first time my fortune hits 10” is a stopping time; “stop one step before the peak” is not. The fair-game intuition suggests no stopping rule can beat a fair game:

**Theorem 8 (Optional stopping (conditions))** If  $(X_n)$  is a martingale and  $T$  a stopping time, then  $E[X_T] = E[X_0]$  provided *at least one* of the following holds:

1.  $T$  is bounded:  $T \leq N$  for a constant  $N$ ; or
2. the increments  $|X_{n+1} - X_n|$  are bounded and  $E[T] < \infty$ ; or
3.  $X_{n \wedge T}$  is dominated by an integrable random variable.

**Common pitfall** The conditions are not decoration: drop them all and the conclusion fails. The famous **doubling (“martingale”) strategy** is the trap.

**Worked example — A counterexample where optional stopping fails.** Take the symmetric random walk  $S_n$  (a martingale,  $S_0 = 0$ ) and stop at the first time it reaches +1:  $T = \inf\{n : S_n = 1\}$ . The walk is recurrent, so  $T < \infty$  almost surely and  $S_T = 1$ . Hence

$$E[S_T] = 1 \neq 0 = E[S_0].$$

There is no contradiction:  $E[T] = \infty$ , the increments-bounded-and-finite-mean condition fails, and so does every other. The “guaranteed” euro is borrowed against an unbounded — and unbounded-time — debt. This is exactly why no betting system beats a fair casino game in practice.

## 4 Random walks

### 4.1 Setup and a sample path

Let  $S_0 = k$  and  $S_n = k + \sum_{i=1}^n \xi_i$ , where the steps  $\xi_i$  are independent,  $\xi_i = +1$  with probability  $p$  and  $\xi_i = -1$  with probability  $q = 1 - p$ . The walk is **symmetric** if  $p = \frac{1}{2}$  and **biased** otherwise.

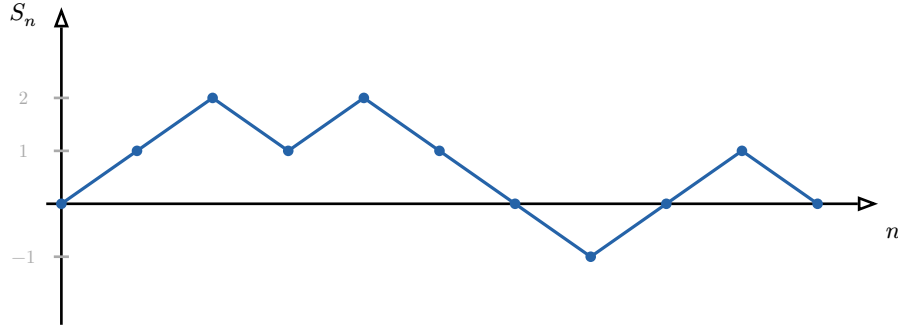


Figure 3: One sample path of a symmetric random walk started at 0, with steps  $\pm 1$ . Each upward segment is a  $+1$ , each downward a  $-1$ .

### 4.2 Recurrence vs. transience

**Proposition 9 (Return behaviour on the integers)** The symmetric walk ( $p = \frac{1}{2}$ ) on  $\mathbb{Z}$  is **recurrent**: started at 0 it returns to 0 with probability 1. The biased walk ( $p \neq \frac{1}{2}$ ) is **transient**: it returns with probability  $< 1$  and drifts to  $+\infty$  (if  $p > \frac{1}{2}$ ) or  $-\infty$  (if  $p < \frac{1}{2}$ ).

**Remark** Recurrence does not mean “returns quickly”. The symmetric walk returns with certainty, yet the *expected* time to return is infinite — the same pathology that broke optional stopping above.

### 4.3 Gambler’s ruin: absorbing barriers

**Worked example — The gambler’s question.** You start with  $k$  euros, bet one euro per fair round, and play until you either go broke (0) or reach a target  $N$ . What is the chance you hit the target first? And how long will you play on average?

The states 0 and  $N$  are **absorbing**: once reached, the walk stays there forever. The interior states  $1, \dots, N-1$  are transient.

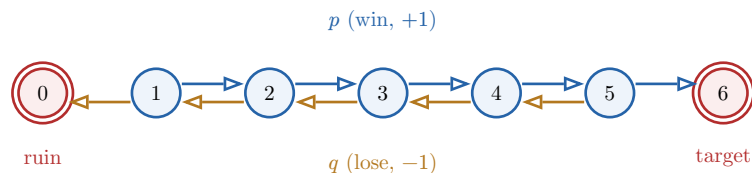


Figure 4: Gambler’s ruin as a walk on  $\{0, \dots, N\}$ . The barriers 0 and  $N$  (double circles) are absorbing; from each interior state the walk steps right with probability  $p$ , left with probability  $q$ .

**Proposition 10 (Ruin / success probabilities)** Let  $h_k = P(\text{reach } N \text{ before } 0 \mid \text{start at } k)$  and  $r = \frac{q}{p}$ . Then

$$h_k = \begin{cases} \frac{k}{N} & \text{if } p = \frac{1}{2} \\ \frac{r^k - 1}{r^N - 1} & \text{if } p \neq \frac{1}{2}. \end{cases}$$

**Proposition 11 (Expected duration (symmetric case))** Let  $\tau_k$  be the expected number of steps until absorption, started at  $k$ . For  $p = \frac{1}{2}$ ,

$$E[\tau_k] = k(N - k).$$

#### 4.4 First-step analysis: the universal technique

Every quantity above is obtained the same way — **first-step analysis**, the probabilistic cousin of the recurrences you solved in MAT11-4. Condition on the very first step:

$$h_k = p h_{k+1} + q h_{k-1}, \quad h_0 = 0, \quad h_N = 1.$$

This is a linear recurrence in  $k$ ; solving it (constant-coefficient, exactly the MAT11-4 method) yields the closed forms above. For the duration, conditioning on the first step costs one move:

$$E[\tau_k] = 1 + p E[\tau_{k+1}] + q E[\tau_{k-1}], \quad E[\tau_0] = E[\tau_N] = 0.$$

**Common pitfall** When  $p = \frac{1}{2}$  the recurrence  $h_k = \frac{1}{2}h_{k+1} + \frac{1}{2}h_{k-1}$  has a *repeated* root, so the solution is *linear* ( $h_k = \frac{k}{N}$ ), not geometric. Blindly writing  $h_k = \frac{r^k - 1}{r^N - 1}$  with  $r = 1$  gives  $\frac{0}{0}$ . Treat the symmetric case separately.

## 5 Markov chains

### 5.1 Memorylessness and the transition matrix

**Worked example — Weather as a chain.** Suppose tomorrow’s weather depends only on today’s, not on the whole past week: if today is sunny, tomorrow is sunny with probability 0.7 and rainy with 0.3; if rainy, sunny with 0.4 and rainy with 0.6. Collect these into a matrix

$$P = \begin{pmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{pmatrix},$$

rows = “today”, columns = “tomorrow”. You already know from MAT11-3 that  $P^n$  gives  $n$ -step transitions; now its entries are *probabilities*.

**Definition 12 (Markov chain and transition matrix)** A sequence  $(X_n)$  on a finite or countable state space  $S$  is a **Markov chain** if it is *memoryless*:

$$P(X_{n+1} = j \mid X_n = i, X_{n-1}, \dots, X_0) = P(X_{n+1} = j \mid X_n = i) =: P_{ij}.$$

The **transition matrix**  $P = (P_{ij})$  has non-negative entries and each row sums to 1 (a *stochastic* matrix). The  $n$ -step probabilities are the entries of  $P^n$ :  $P(X_n = j \mid X_0 = i) = (P^n)_{ij}$ .



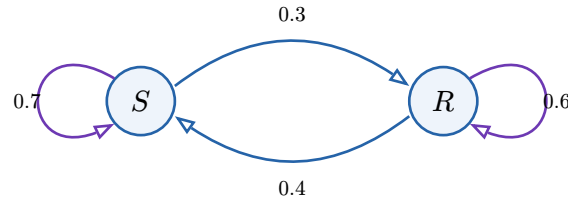


Figure 5: State diagram of the two-state weather chain: nodes are states, arrows carry transition probabilities, and the loops are the “stay” probabilities. Each node’s outgoing probabilities sum to 1.

## 5.2 Classification of states

- A state  $i$  is **absorbing** if  $P_{ii} = 1$  (the chain can never leave).
- $i$  is **recurrent** if, started from  $i$ , the chain returns to  $i$  with probability 1; otherwise it is **transient**.
- The chain is **irreducible** if every state is reachable from every other state (the diagram is strongly connected).
- The **period** of  $i$  is  $\gcd\{n \geq 1 : (P^n)_{ii} > 0\}$ ; the chain is **aperiodic** if every state has period 1. A single self-loop  $P_{ii} > 0$  forces period 1.

## 5.3 Stationary distributions

**Worked example — The long-run weather.** Run the weather chain for a long time: what fraction of days are sunny? Intuition (confirmed below) says the distribution settles to a fixed vector  $\pi$  unchanged by one more day — a row vector with  $\pi P = \pi$ .

**Definition 13 (Stationary distribution)** A row vector  $\pi = (\pi_i)$  with  $\pi_i \geq 0$  and  $\sum_i \pi_i = 1$  is a **stationary distribution** if

$$\pi P = \pi.$$

It is a left eigenvector of  $P$  with eigenvalue 1 (compare the eigenvector language of MAT11-3), normalised to a probability vector.

**Theorem 14 (Convergence to equilibrium)** For an **irreducible, aperiodic** chain on a **finite** state space, a stationary distribution  $\pi$  exists, is unique, and is the limit of the row distributions:

$$P(X_n = j) \rightarrow \pi_j \quad \text{as } n \rightarrow \infty,$$

regardless of the starting state.

**Worked example — Solving the weather chain.** With  $P = \begin{pmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{pmatrix}$ , solve  $\pi P = \pi$  with  $\pi_S + \pi_R = 1$ : from  $0.7\pi_S + 0.4\pi_R = \pi_S$  we get  $0.4\pi_R = 0.3\pi_S$ , so  $\pi_S : \pi_R = 4 : 3$  and

$$\pi = \left( \frac{4}{7}, \frac{3}{7} \right) \approx (0.571, 0.429).$$

In the long run about 57% of days are sunny, whatever today’s weather.

**Common pitfall** Irreducibility and aperiodicity are both required. The chain  $P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  is irreducible but has period 2: started at state 1 it is at 1 on even steps and 2 on odd steps, so  $P(X_n = 1)$  oscillates and never converges, even though  $\pi = (\frac{1}{2}, \frac{1}{2})$  is stationary. Stationary  $\neq$  limiting unless the chain is also aperiodic.

## 6 Concentration inequalities

### 6.1 Turning “mean and variance” into tail bounds

MAT21-2 gave you the mean and variance of an estimator but stopped short of the practical payoff: a *guarantee* that the estimator is close to the truth. Concentration inequalities supply exactly that — they bound the probability of a large deviation using only crude information about  $X$ .

**Theorem 15 (Markov's inequality)** If  $X \geq 0$  and  $a > 0$ , then

$$P(X \geq a) \leq \frac{E[X]}{a}.$$

*Proof.* Since  $X \geq 0$ , the indicator inequality  $a \mathbf{1}_{X \geq a} \leq X$  holds pointwise (on  $X \geq a$  the left side is  $a \leq X$ ; elsewhere it is  $0 \leq X$ ). Taking expectations,  $a P(X \geq a) \leq E[X]$ ; divide by  $a$ .  $\square$

**Theorem 16 (Chebyshev's inequality)** If  $X$  has mean  $\mu$  and finite variance  $\sigma^2$ , then for  $k > 0$ ,

$$P(|X - \mu| \geq k) \leq \frac{\sigma^2}{k^2}.$$

*Proof.* Apply Markov's inequality to the non-negative variable  $(X - \mu)^2$  at threshold  $k^2$ :

$$P(|X - \mu| \geq k) = P((X - \mu)^2 \geq k^2) \leq \frac{E[(X - \mu)^2]}{k^2} = \frac{\sigma^2}{k^2}.$$

$\square$

**Remark** Chebyshev is *derived* from Markov, not postulated — the standard pattern of this chapter: get a sharper bound by feeding Markov a cleverly chosen non-negative function of  $X$ . Hoeffding pushes this to its limit by using the exponential  $e^{\lambda X}$ .

### 6.2 Exponential bounds: Chernoff and Hoeffding

For *sums of bounded independent* variables, polynomial decay (Chebyshev,  $\frac{1}{k^2}$ ) is wasteful: the true tails decay *exponentially*. The **Chernoff method** extracts this by applying Markov to  $e^{\lambda S}$ :

$$P(S \geq a) \leq e^{-\lambda a} E[e^{\lambda S}], \quad \text{then optimise over } \lambda > 0.$$

Carrying it out for bounded summands gives Hoeffding's bound.

**Theorem 17 (Hoeffding's bound)** Let  $X_1, \dots, X_n$  be independent with  $a_i \leq X_i \leq b_i$ , and  $S = \sum_{i=1}^n X_i$ . Then for every  $t > 0$ ,

$$P(S - E[S] \geq t) \leq \exp\left(-\frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right),$$

and the same bound holds for the lower tail  $P(S - E[S] \leq -t)$ .

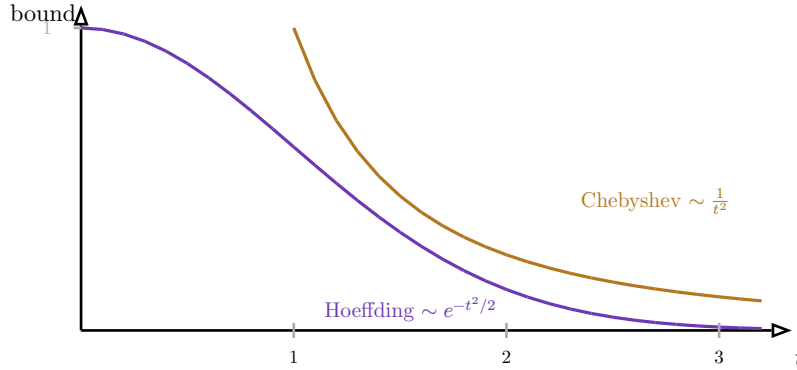


Figure 6: Tail-bound decay against deviation  $t$  (in standard-deviation units). Chebyshev's polynomial  $\frac{1}{t^2}$  (orange) sits far above Hoeffding's exponential  $e^{-t^2/2}$  (purple): both are valid upper bounds, but for sums of bounded variables the exponential bound is dramatically tighter.

**Common pitfall** Hoeffding needs *independence* and *bounded* summands. With heavy tails or strong dependence the exponential bound can be wildly violated; Chebyshev, needing only a finite variance, still holds. Match the inequality to the hypotheses you actually have.

## 7 Quantitative uncertainty control

### 7.1 From a bound to a sample size

The exponential bound pays off as a concrete **sample-size rule**. Let  $X_1, \dots, X_n$  be independent with values in  $[0, 1]$ , common mean  $\mu$ , and sample mean  $\bar{X} = \frac{1}{n} \sum_i X_i$ . Applying Hoeffding to  $S = \sum_i X_i$  (here  $b_i - a_i = 1$ ) and rescaling by  $n$ ,

$$P(|\bar{X} - \mu| \geq \varepsilon) \leq 2 \exp(-2n\varepsilon^2).$$

**Proposition 18 (Minimum sample size)** To guarantee  $P(|\bar{X} - \mu| \geq \varepsilon) \leq \delta$  it suffices to take

$$n \geq \frac{1}{2\varepsilon^2} \ln\left(\frac{2}{\delta}\right).$$

*Proof.* Set the right-hand bound  $2 \exp(-2n\varepsilon^2) \leq \delta$ , take logarithms, and solve for  $n$ :  $2n\varepsilon^2 \geq \ln(\frac{2}{\delta})$ , i.e.  $n \geq \ln(\frac{2}{\delta})/(2\varepsilon^2)$ .  $\square$

This is the “confidence  $1 - \delta$ , tolerance  $\varepsilon$ ” calculation in its purest form. Notice the two scalings: halving the error  $\varepsilon$  costs *four times* the data ( $\frac{1}{\varepsilon^2}$ ), while raising the confidence costs only *logarithmically* in  $\frac{1}{\delta}$  — cheap confidence, expensive precision.

**Worked example — A polling-style computation.** For tolerance  $\varepsilon = 0.05$  and confidence  $1 - \delta = 0.95$  ( $\delta = 0.05$ ),

$$n \geq \frac{1}{2 \cdot 0.05^2} \ln\left(\frac{2}{0.05}\right) = 200 \ln(40) \approx 737.$$

About 740 bounded observations suffice to pin the mean to within 0.05, 19 times out of 20 — *independently* of the underlying distribution.

**Remark** Hoeffding's  $n \approx 737$  is deliberately conservative: it is distribution-free. A normal-approximation calculation (which MAT21-2 hinted at) would ask for fewer observations but assumes a limit that a finite sample only approximates. Concentration trades sharpness for an honest, non-asymptotic guarantee — the right default when you cannot certify normality.

## 8 Python mini-labs

These labs close the loop: implement each model, estimate its tail behaviour by simulation, and hold the estimate up against the theory of the preceding chapters. The point is to *see* recurrence, convergence, and exponential concentration happen, and to feel how loose or tight each bound is.

### Exercises

**1. Simulate a random walk.** Generate  $\pm 1$  steps and accumulate  $S_n$ . Estimate the gambler's-ruin probability  $h_k$  of reaching  $N$  before 0 by Monte Carlo and compare to the closed form  $k/N$  (symmetric) and  $(r^k - 1)/(r^N - 1)$  (biased). Plot a few sample paths like the figure in Chapter 4.

**2. Watch a Markov chain converge.** Iterate the row distribution  $\mu_{n+1} = \mu_n P$  for the weather chain and confirm it approaches  $\pi = (\frac{4}{7}, \frac{3}{7})$  from any start. Then switch to  $P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  and observe that the distribution oscillates and never converges — aperiodicity matters.

**3. Empirically verify concentration.** For a sum  $S$  of  $n$  independent bounded variables, estimate  $P(S - E[S] \geq t)$  by repeated simulation and overlay the observed tail frequency on Hoeffding's bound  $\exp(-2t^2 / \sum (b_i - a_i)^2)$ . Check that the simulated tail stays below the bound.

**4. Compare the bounds.** For a fixed deviation, tabulate the Chebyshev bound, the Hoeffding bound, and the simulated tail probability. Confirm that Hoeffding is exponentially tighter while *both* remain valid upper bounds — the figure in Chapter 7, made empirical.

**5. Stress-test optional stopping.** Simulate the symmetric walk as a martingale. For a bounded stopping time  $T \leq N$ , estimate  $E[S_T]$  and check  $E[S_T] \approx S_0$ . Then take the unbounded doubling-strategy stopping time and watch  $E[S_T]$  drift away from  $S_0$  — optional stopping failing before your eyes.

**6. Sample-size in practice.** Pick a target  $(\varepsilon, \delta)$ , compute the Hoeffding sample size  $n$ , then simulate to estimate the true coverage. Verify the achieved error exceeds  $\varepsilon$  at most a  $\delta$  fraction of the time, and note how much slack the distribution-free bound leaves.